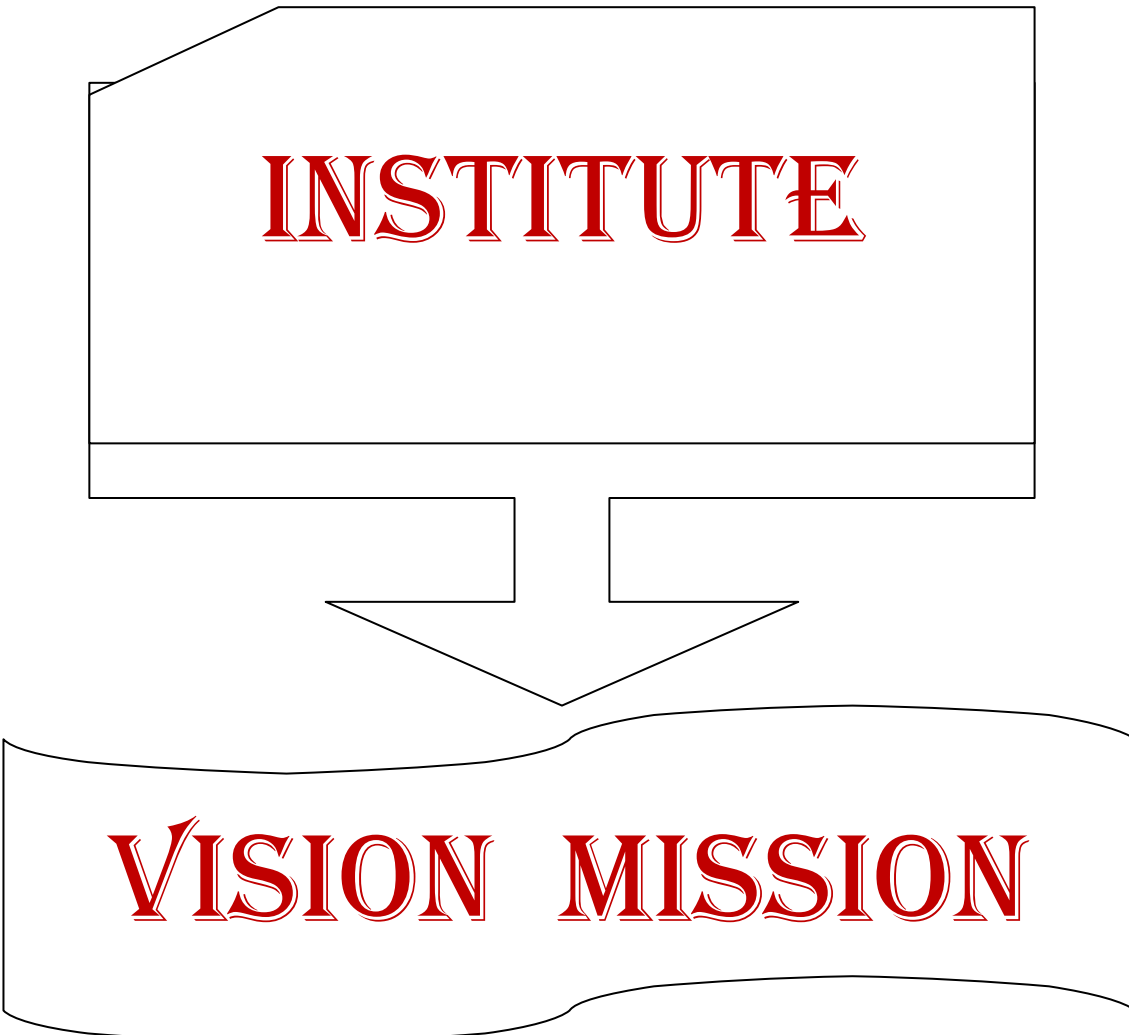


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INSTITUTE

VISION MISSION

INSTITUTE VISION AND MISSION

VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning..



DEPARTMENT



VISION MISSION



DEPARTMENT VISION AND MISSION

Vision:

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission :

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character

**PROGRAM
EDUCATIONAL
OBJECTIVES,
PROGRAM OUTCOMES
& PROGRAM
SPECIFIC
OUTCOMES**

Program Educational Objectives (PEOs)

Graduates of this programme will :

PEO 1: Adapt to evolving technology.

PEO 2: Provide optimal solutions to real time problems.

PEO 3: Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

Programme Specific Outcomes (PSO s):

A graduate of the Computer Science and Engineering Program will be able to:

PSO 1: Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

PSO 2: Use Professional engineering practices and strategies for development and maintenance of software. [K3]

Program Outcomes (POs):

Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. [K3]
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. [K4]
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. [K5]
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. [K5]
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations. [K3]
6. **The Engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. [K3]
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. [K3]
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. [K3]
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. [K6]
10. **Communication:** Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. [K2]
11. **Project management and finance:** Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. [K6]
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. [K1]



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SRI VASAVI ENGINEERING COLLEGE (Autonomous)

(Sponsored by Sri Vasavi Educational Society; Regd.No:898/2000)

| Accredited by NAAC with 'A' Grade | & | Accredited by NBA |

Approved by AICTE, New Delhi and Permanently Affiliated to JNTUK, Kakinada

Pedatadepalli, TADEPALLIGUDEM – 534 101, W.G. Dist, (A.P.)

Dt: 01/07/2020.

Dr. Guduru VNSR Ratnakara Rao

B.E., M.E., Ph.D

PRINCIPAL

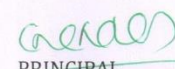
ACADEMIC CALENDAR

For

B.Tech – III Year

2018 Batch

<u>B.Tech V Semester</u>			
<u>Description</u>	<u>From</u>	<u>To</u>	<u>Weeks</u>
Commencement of Class Work	01-07-2020	-	-
I Unit of Instructions	01-07-2020	26-08-2020	8W
I Mid Examination	27-08-2020	05-09-2020	1W
II Unit of Instructions	07-09-2020	31-10-2020	8W
II Mid Examination	02-11-2020	07-11-2020	1W
Comprehensive Test	09-11-2020	14-11-2020	1W
Preparation & Practicals	16-11-2020	21-11-2020	1W
End Examinations	23-11-2020	05-12-2020	2W
Commencement of VI Sem Class work	07-12-2020		


PRINCIPAL

Copy to: All

COURSE STRUCTURE

V-Semester

V - Semester							
S.No.	Course Code		Course	L	T	P	C
1	V18CST10	PCC	Database Management Systems	3	0	0	3
2	V18CST11	PCC	Computer Networks	3	0	0	3
3	V18CST12	PCC	Operating Systems	3	0	0	3
4	V18CST13	PCC	Design and Analysis of Algorithms	3	0	0	3
5	V18CST14	PCC	Unix Programming	3	0	0	3
6	Elective – I						
	V18CST15	PEC	1. Advanced Computer Architecture	3	0	0	3
	V18CST16		2. Advanced Data Structures				
	V18CST17		3. Artificial Intelligence				
	V18CST18		4. Computer Graphics				
7	V18MBET53	HSS	Organizational Behavior	3	0	0	3
8	V18CSL06	PCC	Database Management Systems Lab	0	0	3	1.5
9	V18CSL07	PCC	Operating System and Unix Lab	0	0	3	1.5
10	V18ENT05		Professional Communication Skills –III	4	0	0	MNC
11	V18CST62		Technical Skills-III	4	0	0	MNC
Total Contact Hours: 35				29	0	6	24

Total Contact Hours: 35



LESSON PLANS

DATABASE MANAGEMENT SYSTEMS

Academic Year : 2020-21

Semester : V Sem

Name of the Course: Database Management Systems

Programme: B.Tech

Sections :A,B,C&D

Course Code: V18CST10

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO's	Course Outcome	Knowledge Level
V18CST10.1	Demonstrate Database Systems, various Data Models and Database Architecture.	K2
V18CST10.2	Apply ER Modeling to Design Relational Databases for Real Time Applications.	K3
V18CST10.3	Apply SQL Constructs to Perform Database Operations.	K3
V18CST10.4	Apply Normalization Techniques to Refine Schema.	K3
V18CST10.5	Explain Transaction Management and Concurrency Control.	K2
V18CST10.6	Experiment with various database indexing techniques.	K3

Text Books/ Reference Books suggested:

TEXT BOOKS:

1. Introduction to Database Systems, CJ Date, 8th Edition, Pearson Education.
2. Database Management Systems, Raghurama Krishnan, Johannes Gehrke, 3rd Edition TATA McGraw Hill.

REFERENCE BOOKS:

1. Database Systems design, Implementation, and Management, Peter Rob & Carlos Coronel 7th Edition, Course Technology.
2. Fundamentals of Database Systems, ElmasriNavrate , 7th Edition, Pearson Education.
3. Database Systems - The Complete Book, H G Molina, J D Ullman, J Widom, 2nd Edition, Pearson

Targeted Proficiency Level and Targeted level of Attainment (For each course outcome):

COURSE OUTCOME	PROFICIENCY LEVEL	ATTAINMENT LEVEL
V18CST10.1	65	65
V18CST10.2	65	60
V18CST10.3	60	60
V18CST10.4	60	60
V18CST10.5	60	60
V18CST10.6	65	65

UNIT-I: An Overview of Database Systems

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1		Dissemination of vision, mission, PEOs, POs, PSOs		1	Lecture	PPT
2	CO 1	Define a Database System, Database	K1	1	Lecture	PPT
		Describe the disadvantages in Traditional File System and advantages of DBMS over file system	K1	1	Lecture with Discussion	PPT
3		Describe data independence	K1	1	Lecture with Discussion	PPT
4		Discuss Database Architecture: The Three levels of the architecture: The External level, Conceptual level, Internal level and Mappings	K2	2	Lecture with Discussion	PPT
6		Illustrate Database Management System Architecture and components	K2	1	Lecture with Discussion	PPT
7		Explain different types of users in DBMS	K2	1	Lecture with Discussion	PPT
8		Illustrate the structure of DBMS	K2	1	Lecture with Discussion	PPT
9		Explain Client/Server Architecture	K2	1	Lecture with Discussion	PPT
		TOTAL		10		

UNIT-II: DATABASE DESIGN, RELATIONAL MODEL, RELATIONAL ALGEBRA AND CALCULUS

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Describe introduction to Database Design: Database Design and ER Diagrams	K1	1	Lecture	PPT
2		Explain Entities, Attributes and Entity Sets	K2	1	Lecture With Discussion	PPT
3		Explain Relationships and Relationship Sets	K2	1	Lecture With Discussion	PPT
4		Describe Relational Systems and Others	K2	2	Lecture With Discussion	PPT
5		Illustrate Conceptual Design with the ER Model	K3	1	Lecture With Discussion	PPT
6		Explain The Relational Model: Integrity Constraints over Relations: Key Constraints, Foreign Key Constraints, General Constraints	K2	2	Lecture With Discussion	PPT
7		Illustrate Relational Algebra: Selection and Projection	K3	1	Lecture With Discussion	PPT
8		Illustrate Set Operations, Renaming, Joins, Division, More Example of Algebra Queries	K3	3	Lecture With Discussion	PPT
9		Illustrate Relational Calculus: Tuple Relational Calculus, Domain Relational Calculus	K3	1	Lecture With Discussion	PPT
		TOTAL		13		

UNIT-III: QUERIES, CONSTRAINTS, TRIGGERS

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Explain the Form of a Basic SQL Query	K2	2	Lecture with Demonstration	PPT
2		Illustrate Union, Intersect, Except and Nested Queries	K3	3	Lecture with Demonstration	PPT
3		Illustrate Aggregate Operators	K3	2	Lecture with Demonstration	PPT
4		Interpret Null Values	K2	1	Lecture with Demonstration	PPT
5		Illustrate Complex Integrity Constraints in SQL	K3	1	Lecture with Demonstration	PPT
6		Illustrate Triggers and Active Databases	K3	2	Lecture with Demonstration	PPT
		TOTAL		11		

UNIT-IV: SCHEMA REFINEMENT (NORMALIZATION)

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Explain Purpose of Normalization or Schema Refinement	K2	1	Lecture With Discussion	PPT
2		Explain the Concept of Functional Dependency	K2	2	Lecture With Discussion	PPT
3		Illustrate Normal Forms Based on Functional Dependency (1NF, 2NF and 3 NF)	K3	3	Lecture With Discussion	PPT
4		Demonstrate Concept of Surrogate Key, Boyce- Codd Normal Form (BCNF)	K3	1	Lecture With Discussion	PPT
5		Illustrate Lossless Join and Dependency Preserving Decomposition, Fourth Normal Form (4NF)	K3	2	Lecture With Discussion	PPT
		TOTAL		9		

UNIT-V: TRANSACTION MANAGEMENT AND CONCURRENCY CONTROL

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Describe Transaction, Properties of Transactions, Transaction Log	K2	1	Lecture With Discussion	PPT
2		Explain Transaction Management with SQL Using Commit, Rollback and Save point.	K2	2	Lecture With Discussion	PPT
3		Describe Concurrency Control for Lost Updates, Uncommitted Data, Inconsistent Retrievals and The Scheduler	K2	2	Lecture With Discussion	PPT
4		Explain Concurrency Control with Locking Methods: Lock Granularity, Lock Types, Two Phase Locking for Ensuring Serializability	K2	3	Lecture With Discussion & Seminar	PPT
5		Explain Deadlocks, Concurrency Control with Time Stamp Ordering	K2	2	Lecture With Discussion	PPT
6		Explain Wait/Die and Wound/Wait Schemes	K2	1	Lecture With Discussion	PPT
7		Describe Database Recovery Management: Transaction Recovery	K2	1	Lecture With Discussion	PPT
		TOTAL		12		

UNIT-VI: STORAGE AND INDEXING

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1		Describe Overview of Storages and Indexing	K1	1	Lecture With Discussion	PPT
2	CO 6	Describe data on External Storage	K1	1	Lecture With Discussion	PPT
3		Explain File Organizations and Indexing: Clustered Indexes, Primary and Secondary Indexes	K2	2	Lecture With Discussion	PPT
4		Explain Index Data Structures: Hash-Based Indexing, Tree-Based Indexing	K2	2	Lecture With Discussion	PPT
5		Explain Comparison of File Organizations	K2	2	Lecture With Discussion	PPT
		TOTAL		8		

Total No. of Classes: 63

COMPUTER NETWORKS

Academic Year : 2020-21

Semester : V Sem

Name of the Course: Computer Networks

Programme: B.Tech

Sections :A,B,C&D

Course Code: V18CST11

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

- CO1: Discuss fundamentals of network concepts and Reference Models (K2)
CO2: Discuss Communication media and switching techniques. (K2)
CO3: Demonstrate Error control and protocols. (K3)
CO4: Apply Routing algorithms and congestion control algorithms. (K3)
CO5: Discuss Transport layer services and protocols. (K2)
CO6: Describe Application layer protocols. (K2)

Text Books:

1. Computer Networks — Andrew S Tanenbaum, 4th Edition. Pearson Education/PHI
2. Data Communications and Networks – Behrouz A. Forouzan, Third Edition TMH

Reference Books:

1. An Engineering Approach to Computer Networks-S.Keshav, 2nd Ed, Pearson Education
 2. Understanding communications and Networks, 3rd Edition, W.A. Shay, Thomson
- Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Targeted Proficiency Level (For each course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)
1	70
2	65
3	65
4	60
5	60
6	60

Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted level of Attainment (% Students)
1	70
2	60

3	60
4	60
5	60
6	60

CO 1

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of Vision, Mission of the Dept. and PEOs, Pos,& PSOs of the Programme		1	Lecture	BB+ICT
2		Describe Reference models- The OSI Reference Model	K1	2	Lecture	BB+ICT
3		Discuss TCP/IP Reference Model	K2	1	Lecture with Discussion	BB+ICT
4		Examples of Networks: ARPANET, INTERNET	K2	1	Lecture	BB+ICT
5		Describe Network Topologies WAN, LAN, MAN.	K1	1	Lecture	BB+ICT

CO 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Describe Physical Layer – Fourier Analysis, The Maximum Data	K1	1	Lecture	BB+ICT
2		Explain Different Kinds of Transmission Media	K2	2	Lecture	BB+ICT
3		Describe Multiplexing Techniques: Frequency Division Multiplexing and Wavelength Division Multiplexing	K1	2	Lecture	BB+ICT
4		Discuss Time Division Multiplexing	K1	1	Lecture with Discussion	BB+ICT
5		Explain Circuit Switched Networks	K2	1	Lecture with Discussion	BB+ICT
6		Explain Datagram Networks and Virtual Circuit Networks	K2	2	Lecture with Discussion in class Assignment	BB+ICT

CO 3

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Describe The Data Link Layer - Services Provided to the Network Layer	K1	1	Lecture	BB+ICT
2		Explain Data Link Layer Design Issues	K2	1	Lecture	BB+ICT
3		Illustrate Framing – Error Control –Flow Control, Error Detection and Correction – Error-Correcting Codes – Error Detecting code	K3	2	Lecture with Discussion and in class Assignment	BB+ICT
4		Discuss MAC Protocols: ALOHA, CSMA	K2	2	Lecture with Discussion	BB+ICT
5		Explain Elementary Data Link Protocols- A Utopian Simplex Protocol	K2	1	Lecture with Discussion	BB+ICT
6		Discuss A Simplex Stop and Wait Protocol for an Error free channel-A Simplex Stop and Wait Protocol for a Noisy Channel.	K2	1	Lecture	BB+ICT
7		Demonstrate Sliding Window Protocols	K3	3	Lecture with Discussion	BB+ICT
8		Discuss HDLC, PPP and Piggybacking	K2	2	Lecture	BB+ICT

CO 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Describe The Network Layer Design Issues Store-and-Forward Switching, Services Provided to Transport Layer	K1	2	Lecture	BB+ICT
2		Discuss Implementation of Connectionless Service-Implementation of Connection Oriented Service.	K2	2	Lecture with Discussion	BB+ICT
3		Construct Shortest Path algorithm	K3	1	Lecture with Discussion	BB+ICT
4		Describe Flooding	K2	1	Lecture	BB+ICT

5		Illustrate Distance Vector Routing and Hierarchical Routing	K3	2	Lecture and in class Assignment	BB+ICT
6		Differentiate Broadcast and Multicast Routing	K2	1	Lecture	BB+ICT
7		Discuss Congestion Control Algorithms	K2	1	Lecture with Discussion	BB+ICT
8		Demonstrate IP Addressing and Subnet Masking	K3	1	Lecture with Discussion and in class Assignment	BB+ICT

CO 5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Describe Transport Layer – Services, Primitives and Sockets	K1	2	Lecture	BB+ICT
2		Discuss Elements of Transport Protocols	K2	2	Lecture	BB+ICT
3		Discuss The Internet Transport Protocols: TCP Segment Header and Primitives	K2	2	Lecture with Discussion	BB+ICT
4		Discuss The Internet Transport Protocols: UDP, RPC, RTP, and RTCP Segment Headers and Primitives	K2	1	Lecture with Discussion	BB+ICT
5		Discuss Congestion Control in TCP	K2	1	Lecture with Discussion	BB+ICT
6		Outline Quality of Service	K1	1	Lecture	BB+ICT
7		Discuss QoS Improving Techniques: Leaky Bucket and Token Bucket Algorithms	K2	1	Lecture with Discussion	BB+ICT

CO 6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Discuss Application Layer – DNS: The DNS Name Space, Resource Records, Name Servers.	K2	1	Lecture with Discussion	BB+ICT
2		Describe Electronic Mail Message Transfer Agent: SMTP and Message Access Agent: POP	K2	2	Lecture	BB+ICT
3		Discuss FTP: Control Connection and Data Connection	K2	1	Lecture with Discussion	BB+ICT
		Discuss HTTP: Connections, Methods, Message Headers, and Caching	K2	1	Lecture with Discussion	BB+ICT
4		Describe Cryptography and Authentication	K2	1	Lecture	BB+ICT
5		Differentiate Public Key and Private Key Cryptographic Algorithms	K2	2	Lecture with Discussion	BB+ICT
6		Discuss Digital Signatures and Certificates	K2	1	Lecture with Discussion	BB+ICT
7		Outline the working of Firewalls	K2	1	Lecture	BB+ICT
8		Discuss Wireless Security	K2	1	Lecture	BB+ICT

Total No. of Classes: 60

OPERATING SYSTEM

Academic Year: 2020-21

Programme: B.Tech

Year/ Semester: V Sem

Section: A, B, C & D

Name of the Course: Operating System

Course Code: : V18CST12

After completion of this course, the student will be able to:

CO1	Describe Operating System Services and System Calls	K2
CO2	Illustrate Process Management Concepts and CPU Scheduling Algorithms	K3
CO3	Demonstrate Process Synchronization primitives	K3
CO4	Demonstrate Deadlock Prevention, Avoidance and Detection methods	K3
CO5	Illustrate Memory Management Techniques and Page Replacement Algorithms	K3
CO6	Describe File System Concepts and Mass Storage Structures	K2

Text Books:

1. Operating System Concepts, Abraham Silberschatz, ,Peter Baer Galvin, Greg Gagne, 9th Edition, John Wiley and Sons Inc., 2012

Reference Books:

1. Operating Systems – Internals and Design Principles, William Stallings, 7th Edition, Prentice Hall, 2012
2. Modern Operating Systems, Andrew S. Tanenbaum, Third Edition, Addison Wesley, 2007

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		65	65	65	60	60	60
Targeted level of Attainment	Level 3	65	65	65	60	60	60
	Level 2	55	55	55	50	50	50
	Level 1	45	45	45	40	40	40

Lecture Plan:

S N O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teachin g aids
1	CO 1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs	-	-	-	
		Introduction of OS	K2	1	Lecture with Discussion	BB/ICT
2		Operating-System Structure	K2	1	Lecture with Discussion	BB/ICT
3		Operating-System Services	K2	2	Lecture with Discussion	BB/ICT
4		User and Operating-System Interface	K2	1	Lecture with Discussion	BB/ICT
5		System Calls	K2	1	Lecture with Discussion	BB/ICT
6		Types of System Calls	K2	1	Lecture with Discussion	BB/ICT

SN O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO2	Process Scheduling	K3	2	Lecture with Discussion and in class Assignment	ICT
2		Operations On Processes	K3	1	Lecture with Discussion and in class Assignment	ICT
3		Inter Process Communication(IPC)	K3	1	Lecture with Discussion and in class Assignment	ICT
4		Threads Introduction	K3	1	Lecture with Discussion and in class Assignment	ICT
5		Multithreading Models	K3	1	Lecture with Discussion and in class Assignment	ICT
6		CPU Scheduling Basic Concepts	K3	1	Lecture with Discussion and in class Assignment	ICT
7		CPU Scheduling Criteria	K3	1	Lecture with Discussion and in class Assignment	ICT
8		CPU Scheduling Algorithms	K3	4	Lecture with Discussion and in class Assignment	ICT

S N O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO3	Critical Section Problem	K3	1	Lecture with Discussion and in class Assignment	ICT
2		Peterson's Solution	K3	1	Lecture with Discussion and in class Assignment	ICT
3		Synchronization Hardware			Lecture with Discussion and	ICT

			K3	1	in class Assignment	
4		Mutex Locks	K3	2	Lecture with Discussion and in class Assignment	ICT
5		Semaphores	K3	1	Lecture with Discussion and in class Assignment	ICT
6		Classic Problems of Synchronization	K3	2	Lecture with Discussion and in class Assignment	ICT
7		Monitors	K3	2	Lecture with Discussion and in class Assignment	ICT

S N O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO4	Dead Lock Introduction	K3	1	Lecture with Discussion and in class Assignment	ICT
2		System Model	K3	1	Lecture with Discussion and in class Assignment	ICT
3		Deadlock Characterization	K3	1	Lecture with Discussion and in class Assignment	ICT
4		Methods for Handling Deadlocks,	K3	2	Lecture with Discussion and in class Assignment	ICT
5		Deadlock Prevention	K3	2	Lecture with Discussion and in class Assignment	ICT
6		Deadlock Avoidance	K3	1	Lecture with Discussion and in class Assignment	ICT
7		Deadlock Detection	K3	2	Lecture with Discussion and in class Assignment	ICT

8		Recovery from Deadlock	K3	2	Lecture with Discussion and in class Assignment	ICT
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S N O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO5	Swapping	K3	1	Lecture with Discussion and in class Assignment	ICT
2		Contiguous Memory Allocation	K3	1	Lecture with Discussion and in class Assignment	ICT
3		Segmentation,	K3	1	Lecture with Discussion and in class Assignment	ICT
4		Paging,	K3	1	Lecture with Discussion and in class Assignment	ICT
5		Structure of the Page Table	K3	1	Lecture with Discussion and in class Assignment	ICT
6		Demand Paging	K3	1	Lecture with Discussion and in class Assignment	ICT
7		Page Replacement	K3	3	Lecture with Discussion and in class Assignment	ICT
8		Allocation of Frames	K3	1	Lecture with Discussion and in class Assignment	ICT
9		Thrashing	K3	1	Lecture with Discussion and in class Assignment	ICT

S N O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO6	Overview of Mass-Storage Structure	K2	1	Lecture with Discussion	BB/ICT
2		Disk Scheduling	K2	2	Lecture with Discussion	BB/ICT
3		File Concept, Access Methods	K2	1	Lecture with Discussion	BB/ICT
4		Directory and Disk Structure	K2	1	Lecture with Discussion	BB/ICT
5		File-System Mounting	K2	1	Lecture with Discussion	BB/ICT
6		File Allocation Methods	K2	1	Lecture with Discussion	BB/ICT

Total Number of Hours: 60

DESIGN AND ANALYSIS OF ALGORITHMS

Academic Year: 2020-21

Year/ Semester: V Semester

Name of the Course: Design and Analysis of Algorithms

Programme: B.Tech

Section: A,B,C & D

Course Code: V18CST13

COURSE OUTCOMES (Along with Knowledge Level):

After successful completion of the Course, the student will be able to:

After the completion of this course, students will be able to:

V18CST13.1 - Describe asymptotic notations and basic concepts of algorithms (**K2**)

V18CST13.2 - Apply divide and conquer paradigm to solve various problems (**K3**)

V18CST13.3 - Use greedy technique to solve various problems (**K3**)

V18CST13.4 - Apply dynamic programming technique to various problems (**K3**)

V18CST13.5 - Employ backtracking technique to various problems (**K3**)

V18CST13.6 - Apply branch and bound technique to various problems (**K3**)

TEXT BOOKS:

- Fundamentals of Computer Algorithms - Ellis Horowitz, Satraj Sahni and Rajasekaran; 2/e; Universities Press.
- Design and Analysis of Algorithms - S Sridhar; Oxford.

REFERENCE BOOKS:

- Design and Analysis of Algorithms - Aho, Ullman and Hopcroft; Pearson Education.
- Introduction to Algorithms - T.H.Cormen, C.E.Leiserson, R.L.Rivest and C. Stein; 2/e; PHI

TARGETED PROFICIENCY AND ATTAINMENT LEVELS:

Course Outcome	Targeted Proficiency (Marks In %)	Targeted Attainment (Students In %)
V18CST13.1	60	70
V18CST13.2	60	70
V18CST13.3	60	70
V18CST13.4	60	70
V18CST13.5	60	70
V18CST13.6	60	70

LESSON PLAN

UNIT - 1: Introduction						
S. No.	Course Outcome	Intended Learning Outcome (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aid
1	V18CST13.1	What is an Algorithm, Properties of Algorithm	K1	01	Lecture	BB
2		Algorithm Specification- Pseudo Code Conventions, Recursive Algorithms	K2	01	Lecture	BB
3		Explain Performance Analysis-Space Complexity, Time Complexity	K2	03	Lecture	BB
4		Describe Asymptotic Notations- Big oh notation, Omega notation, Theta notation and Little oh notation	K2	01	Lecture	BB
5		Practical Complexities	K2	02	Lecture	BB
6		Explain Amortized Analysis, Probabilistic Analysis	K2	01	Lecture	BB
7		Performance Measurement	K2	01	Lecture	BB
			Total:	10		

UNIT - 2: Divide and Conquer						
S. No.	Course Outcome	Intended Learning Outcome (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aid
1	V18CST13.2	Describe Divide and Conquer strategy ,Control Abstraction	K1	01	Lecture	BB
2		Applications –Finding the minimum and maximum	K2	01	Lecture	BB
3		Apply Divide and Conquer to Binary Search	K3	01	Lecture + Discussion	BB
4		Apply Divide and Conquer to Quick Sort	K3	02	Lecture + Discussion	BB
5		Apply Divide and Conquer to Merge Sort, Refinements to merge sort	K3	02	Lecture + Discussion	BB
			Total:	07		

UNIT - 3: Greedy Method						
S. No.	Course Outcome	Intended Learning Outcome (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aid
1	V18CST13.3	Describe Greedy method, Control Abstraction	K1	01	Lecture	BB
2		Applications - Solve Job Sequencing with deadlines	K3	02	Lecture + Discussion	BB
3		Solve Knapsack Problem	K3	02	Lecture + Discussion	BB
4		Explain Spanning Trees, Find Minimum Cost Spanning Trees(Prim's and Kruskal's algorithms)	K3	02	Lecture + Discussion	BB
5		Solve Single Source Shortest Path Problem	K3	01	Lecture + Discussion	BB
6		Optimal Merge Patterns	K3	01	Lecture + Discussion	BB
			Total:	09		

UNIT - 4: Dynamic Programming						
S. No.	Course Outcome	Intended Learning Outcome (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aid
1	V18CST13.4	Describe Dynamic Programming	K1	01	Lecture	BB
2		Applications - Solve All Pairs Shortest Path Problem	K3	01	Lecture + Discussion	BB
3		Single Source shortest Paths General Weights	K3	01	Lecture + Discussion	BB
4		Explain Optimal Binary Search Trees	K3	03	Lecture + Discussion	BB
5		String Edition	K3	02	Lecture + Discussion	BB
6		Solve 0/1 Knapsack Problem	K3	01	Lecture + Discussion	BB
7		Explain Reliability Design	K3	03	Lecture + Discussion	BB
			Total:	12		

UNIT - 5: Backtracking						
S. No.	Course Outcome	Intended Learning Outcome (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aid
1	V18CST13.5	Explain General method	K2	01	Lecture	BB
2		Applications: Solve N-Queen Problem	K3	02	Lecture + Discussion	BB
3		Employ Backtracking to solve Sum of Subsets Problem	K3	03	Lecture + Discussion	BB
4		Demonstrate Graph Coloring	K3	02	Lecture + Discussion	BB
5		Demonstrate Hamiltonian Cycles	K3	02	Lecture + Discussion	BB
			Total:	10		

UNIT - 6: Branch and Bound						
S. No.	Course Outcome	Intended Learning Outcome (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching Aid
1	V18CST13.6	General method of Branch and Bound technique	K1	01	Lecture	BB
2		Applications - Solve Travelling Sales Person problem	K3	02	Lecture + Discussion	BB
3		Apply Branch and Bound to 0/1 Knapsack problem	K3	03	Lecture + Discussion	BB
4		Explain LC Branch and Bound solution	K3	03	Lecture + Discussion	BB
5		Explain FIFO Branch and Bound solution	K3	03	Lecture + Discussion	BB
			Total:	12		

Total Number of Hours: 60

UNIX PROGRAMMING

Academic Year: 2020-21

Year/ Semester: V Sem

Name of the Course: Unix Programming

Course Outcomes (Along with Knowledge Level):

After Completing the course Student will be able to:

Programme: B.Tech

Section: A, B, C & D

Course Code: : V18CST14

CO	Course Outcomes	Knowledge Level
1	Illustrate the UNIX basics and the working of the built in commands in Unix	K2
2	Demonstrate the file system and change the permissions associated with files	K2
3	Develop basic programs using shell script	K3
4	Demonstrate the grep family and data transforming programs sed, and awk	K2
5	Construct programs for process system calls	K3
6	Explain the concept of signals and its system call	K2

Text Books:

1. Introduction to Unix and shell programming, M G venkatesh murthy, Pearson education
Advanced programming in the unix environment, W. Richard Stevens, 3rd Edition, Pearson education.

REFERENCES

1. Unix and shell Programming, B.A. Forouzan& R.F. Giberg, ,Thomson, First Edition, New Delhi, 2003.

Targeted Proficiency and Attainment Levels (for each course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60	60
	Level 2	50	50	50	50	50	50
	Level 1	40	40	40	40	40	40

Lecture Plan:**UNIT – I**

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs, COs, Introduction to Unix Programming and Describe Brief History	K1	1	Lecture with discussion	PPT
2.		Explain The UNIX Architecture and Basic features of UNIX	K2	1	Lecture with discussion	PPT
3.		Demonstrate General Purpose Utilities:cal, date, man, echo, bc, clear, passwd, who, whoami, uname and Directory Handling commands:pwd, cd, mkdir, rmdir	K2	1	Lecture with demonstration	PPT, CLI
4.		Demonstrate File Handling Utilities: cat, touch, cp, ls, rm, mv, nl, pg,tar,wc	K2	2	Lecture with demonstration	PPT, CLI
5.		Demonstrate Displaying Commands:more,head, tail and simple filter commands:cmp, comm., ulink, diff, head, tail, find, cut, paste,sort, uniq, tr, finger	K2	2	Lecture with demonstration	PPT, CLI
6.		Demonstrate Disk Utilities: du, df, mount, umount and Process Utilities: ps, kill	K2	1	Lecture with demonstration	PPT, CLI
7.		Demonstrate Networking Utilities–ping, telnet, rlogin, ftp	K2	1	Lecture with demonstration	PPT, CLI
Total Hours				9 Hours		

UNIT – II

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO2	Discuss the Types of Files, Directories and Files,	K2	1	Lecture with discussion	PPT
2.		Explain the UNIX File System and Absolute and relative pathnames	K2	1	Lecture with discussion	PPT
3.		Discuss the File Attributes and Permissions	K2	1	Lecture with discussion	PPT
4.		Discuss File Command knowingthe File Type	K2	1	Lecture with discussion	PPT
5.		Demonstrate Chmod Command Changing File Permissions	K2	2	Lecture with demonstration	PPT, CLI
6.		Demonstrate Chown Command Changing the Owner of a File and Chgrp Command Changing the Group of a File.	K2	1	Lecture with demonstration	PPT, CLI
7.		Explain Vi editor- editing with vi, moving the cursor, editing, copying and moving text, pattern searching.	K2	1	Lecture with demonstration	PPT, Live Demo
Total Hours				8 Hours		

UNIT – III

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO3	Explain Shell Variables	K2	2	Lecture with discussion	PPT
2.		Examine The Export Command and Explain The .Profile File a Script Run During Starting	K2	1	Lecture with discussion	PPT
4.		Demonstrate The First Shell Script and read Command	K3	1	Lecture with demonstration	PPT
5.		Explain the Positional parameters and Explain The \$?Variable knowing the exit Status	K2	2	Lecture with discussion	PPT

6.		Demonstrate More about the Set Command and Practice The Exit Command	K3	1	Lecture with demonstration	PPT
7.		Practice on Branching Control Structures	K3	3	Lecture with demonstration	PPT
8.		Practice on Loop Control Structures	K3	2	Lecture with demonstration	PPT
9.		Examine The Continue and Break Statement	K3	1	Lecture with demonstration	PPT
10.		Demonstrate Expr Command: Performing Integer Arithmetic and Real Arithmetic in Shell programs	K3	2	Lecture with demonstration	PPT
11.		Explain The here document(<<) and Demonstrate the Sleep Command	K2	1	Lecture with demonstration	PPT
12.		Demonstrate Debugging Scripts and Demonstrate the Eval Command	K3	1	Lecture with demonstration	PPT
13.		Explain The Exec Command.	K2	1	Lecture with demonstration	PPT
14.		Illustrate Command Line Structure	K3	1	Lecture with discussion	PPT
15.		Demonstrate Meta characters	K3	1	Lecture with demonstration	PPT
Total Hours				20 Hours		

UNIT – IV

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO4	Demonstrate the Grep Family-egrep, fgrep	K2	2	Lecture with demonstration	PPT, CLI
2.		Demonstrate the Stream Editor Sed	K2	2	Lecture with	PPT

					demonstration	
3.		Demonstrate The AWK Pattern Scanning and processing Language	K2	3	Lecture with demonstration	PPT
Total Hours				07 Hours		

UNIT – V

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO5	Explain Process and its structure	K2	1	Lecture with discussion	PPT
2.		Explain the Process Identifiers	K2	1	Lecture with discussion	PPT
3.		Demonstrate Process Creation – fork(),V fork() ,exit() System calls	K3	1	Lecture with demonstration	PPT
4.		Demonstrate waiting for a process – wait(), wait pid() System calls	K3	1	Lecture with demonstration	PPT
5.		Explain Zombie Process	K2	1	Lecture with discussion	PPT
6.		Demonstration of exec system call.	K3	1	Lecture with demonstration	PPT
Total Hours				06 Hours		

UNIT – VI

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO6	Explain Signal and signal function	K2	1	Lecture with discussion	PPT
2.		Explain unreliable signals	K2	1	Lecture with discussion	PPT

3.		Explain Interrupted system calls	K2	1	Lecture with discussion	PPT
4.		Explain kill and raise Functions	K2	2	Lecture with demonstration	PPT
5.		Explain alarm, pause functions, abort, sleep functions	K2	3	Lecture with demonstration	PPT
Total Hours				08Hours		

Total No. of classes: 58

ARTIFICIAL INTELLIGENCE

Academic Year: 2020-21

Programme: B.Tech

Semester : V Sem

Section: A, B, C&D

Name of the Course: Artificial Intelligence

Course Code: V18CST17

Course Outcomes (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO	Course Outcomes	Knowledge Level
1	Illustrate the concept of Intelligent systems and current trends in AI	K2
2	Apply Problem solving, Problem reduction and Game Playing techniques in AI	K3
3	Illustrate the Logic concepts in AI	K2
4	Explain the Knowledge representation techniques in AI	K2
5	Describe Expert systems and their applications	K2
6	Illustrate Uncertainty Measures	K2

TEXT BOOKS:

1. Artificial Intelligence, Saroj Kaushik, 1st Edition, Cengage Learning

REFERENCE BOOKS:

1. Artificial Intelligence, Elaine Rich, Kevin Knight, Shiva shankar B Nair, 3rd Edition, Tata McGraw Hill Education Private Limited., 2009
2. Artificial Intelligence- A modern Approach, 3rd Edition, Stuart Russel, Peter Norvig, Pearson Education

Targeted Proficiency and Attainment Levels (for each course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60	60
	Level 2	50	50	50	50	50	50
	Level 1	40	40	40	40	40	40

Lecture Plan:

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO 1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs	-	-	-	
		Explain AI Introduction and history	K1	1	Lecture	ICT
2		Describe Intelligent Systems	K2	1	Lecture With Discussion	ICT
3		Discuss foundations of AI	K2	1	Lecture With Discussion	ICT
4		Describe AI applications	K2	1	Lecture With Discussion	ICT
5		Illustrate Tic-Tac-Toe game playing approaches	K2	2	Lecture With Discussion	ICT
6		Describe current trends in AI	K2	1	Lecture With Discussion	ICT

07

S.No	Course Outcome	Intended Learning Outcomes(ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Illustrate general problem solving approaches and problem characteristics	K3	2	Lecture with Discussion and in class Assignment	ICT
2		Demonstrate exhaustive searches	K3	2	Lecture with Discussion and in class Assignment	ICT
3		Demonstrate Heuristic search techniques	K3	3	Lecture with Discussion and in class Assignment	ICT
4		Illustrate Iterative Deepening A* and constraint satisfaction	K3	2	Lecture with Discussion and in class Assignment	ICT
5		Demonstrate problem reduction approaches	K3	1	Lecture with Discussion and in class	ICT

					Assignment	
6		Demonstrate Game playing approaches	K3	2	Lecture with Discussion and in class Assignment	ICT
7		Use Bounded look-ahead strategy, evaluation functions and MINIMAX procedure	K3	1	Lecture with Discussion and in class Assignment	ICT
8		Illustrate Alpha-Beta pruning and two player perfect information games	K3	1	Lecture with Discussion and in class Assignment	ICT

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S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Explain Propositional calculus	K2	1	Lecture with Discussion	BB/ICT
2		Demonstrate Propositional Logic	K2	1	Lecture with Discussion	BB/ICT
3		Demonstrate Natural Deduction system	K2	1	Lecture with Discussion	BB/ICT
4		Explain Axiomatic system	K2	1	Lecture with Discussion	BB/ICT
5		Demonstrate semantic tableau system in propositional logic	K2	1	Lecture with Discussion	BB/ICT
6		Explain resolution refutation in propositional logic	K2	2	Lecture with Discussion	BB/ICT
7		Demonstrate predicate logic	K2	3	Lecture with Discussion	BB/ICT

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S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Explain approaches to knowledge representation	K2	1	Lecture with Discussion	BB/ICT
2		Demonstrate Knowledge Representation(KR) using semantic network	K2	2	Lecture with Discussion	BB/ICT
3		Demonstrate Extended semantic networks for KR	K2	2	Lecture with Discussion	BB/ICT

4		Describe Knowledge Representation using Frames	K2	2	Lecture with Discussion	BB/ICT
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07

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Discuss phases in building expert systems	K2	2	Lecture with Discussion	BB/ICT
2		Explain Expert systems versus Traditional systems	K2	2	Lecture with Discussion	BB/ICT
3		Discuss Rule-based Expert systems	K2	2	Lecture with Discussion	BB/ICT
4		Explain Blackboard systems	K2	2	Lecture with Discussion	BB/ICT
5		Explain Truth Maintenance Systems	K2	2	Lecture with Discussion	BB/ICT
6		List applications of expert systems	K2	1	Lecture with Discussion	BB/ICT

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S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Explain Probability Theory	K2	3	Lecture with Discussion	BB/ICT
2		Explain Bayesian Belief Networks	K2	2	Lecture with Discussion	BB/ICT
3		Discuss Certainty Factor theory	K2	1	Lecture with Discussion	BB/ICT
4		Discuss Dempster-Shafer theory	K2	2	Lecture with Discussion	BB/ICT

08

Total No. of Classes: 57

Organizational Behaviour

Academic Year: 2020-21

Programme: B.Tech

Semester : V Sem

Section: A, B, C&D

Name of the Course: Organizational Behaviour

Course Code: V18MBT53

Course Outcomes (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO	K _x	Course Outcome
V18MBT53.1	K ₁	Identify the basic concepts of organizational behaviour
V18MBT53.2	K ₂	Enumerate the importance of individual personality and learning in the organizational context.
V18MBT53.3	K ₃	Develop decision making abilities and interpersonal communication skills
V18MBT53.4	K ₁	Identify the basic concepts of Group dynamics
V18MBT53.5	K ₁	Identify the foundations of organization development
V18MBT53.6	K ₂	Develop team building skills

Text Books/ Reference Books suggested:

1. K. Aswathappa: "Organizational Behavior-Text, Cases and Games", Himalaya Publishing House, New Delhi, 2008.
2. Jennifer M.George and Gareth R. Jones: "Understanding and Managing organizational Behavior", Pearson Education, New Delhi, 2009.
3. Pareek Udai: "Understanding Organizational Behavior", Oxford University Press, New Delhi, 2007

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	60	60
2	60	60
3	60	60
4	60	60
5	60	60
6	60	60

S. no	Course Outcome	Contents	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids	Remarks
	Identify the basic concepts of OB	UNIT–Introduction to Organizational Behaviour		9			
1		Define OB, nature and scope	K1	2	Lecture Discuss	Blackboard	
2		Describe OB linkages with other social science	K1	1	Lecture	Blackboard	
3		Explain Individual and organizational goals	K2	1	Lecture	Blackboard	
4		Describes perspectives of human behaviour	K1	1	Lecture Discuss	Blackboard	
5		Describe approaches to OB	K2	2	Lecture	Blackboard	
6		Explain Models of OB	K2	2	Lecture	Blackboard	
	Enumerate the importance of individual personality and learning in the organizational context.	UNIT–II: Personality Development		12			
1		Define personality, nature, stages in personality development	K1	1	Lecture	Blackboard	
2		Describe determinants of personality	K2	1	Lecture	Blackboard	
3		Johari Window	K2	1	Lecture	Blackboard	
4		Transactional Analysis	K2	2	Lecture Discuss	Blackboard	
5		Describe learning processes	K1	1	Lecture	Blackboard	
		Explain learning theories	K2	1	Lecture	Blackboard	
		Describe perception, nature and process	K2	1	Lecture	Blackboard	
6		Motivation concepts	K2	1	Lecture	Blackboard	
7		Motivation theories	K2	1	Lecture	Blackboard	
8		Leadership theories	K2	1	Lecture	Blackboard	

9		Explain attitude and values	K1	1	Lecture	Blackboard	
		UNIT–III: Decision making process		5			
1	Develop decision making abilities and interpersonal communication skills	Describe Decision making process	K1	1	Lecture	Blackboard	
2		Explain Individual Vs team decision making	K2	1	Lecture	Blackboard	
3		Describe Interpersonal communication	K2	1	Lecture	Blackboard	
4		Describe Types of decisions	K2	2	Lecture	Blackboard	
		UNIT–IV: Group Dynamics		10			
1	Identify the basic concepts of Group dynamics	Explain Behavioural dimension of groups	K1	1	Lecture	Blackboard	
2		Explain Group formation, informal organization	K1	2	Lecture	Blackboard	
3		Describe group vs. individual interaction.	K1	1	Lecture	Blackboard	
4		Explain group conflict	K1	1	Lecture	Blackboard	
5		Listening and feedback in groups	K2	1	Lecture	Blackboard	
6		Explain collaboration in groups	K1	1	Lecture	Blackboard	
7		Describe conflict resolution in groups	K2	2	Lecture	Blackboard	
8		Explain problem solving techniques in groups	K1	1	Lecture	Blackboard	
		UNIT–V: Organizational Development		10			
1	Identify the foundations of organization development	Explain OD, nature, goals and processes of OD	K1	1	Lecture	Blackboard	
2		Explain Change and resistance to change in OD	K1	1	Lecture	Blackboard	
3		Describe the impact of technology on human behaviour.	K2	1	Lecture	Blackboard	
4		OD interventions	K2	3	Lecture	Blackboard	
5		OD techniques	K2	3	Lecture	Blackboard	

6		Explain OD applications	K1	1	Lecture	Blackboard	
		UNIT–VI: Team Building		07			
1	Develop team building skills	Explain meaning and significance of team building	K1	2	Lecture	Blackboard	
2		Explain stages of team building	K2	2	Lecture	Blackboard	
3		Describe benefits of team building	K1	1	Lecture	Blackboard	
4		Explain formation of high performance teams	K2	2	Lecture	Blackboard	

Total No. of Classes: 53

Professional Communication Skills - III

Academic Year : 2020-21

Semester: II/IV

Branch: CSE

Name of the Course: Professional Communication Skillss-III

Programme:B.Tech

Sections:A,B,C& D

Course Code: V18ENT05

COURSE OUTCOMES:

Course Outcomes (Along with Knowledge Level):

After completing this course the student able to:

- CO1: Distinguish the subtle meanings of various words in different contexts, recognize similar words as well as words with contrast meanings and use them appropriately.[K2]
- CO2: Interpret the passage using different strategies and answer the questions with ease.[K3]
- CO3: Compare different pairs of words and draw analogy between the words. Choose an appropriate word to make a sentence meaningful.[k4]
- CO4: Recognize the easiest and best possible way of solving problem in the area of Number and Letter Series, Analogy, Classification, Coding & Decoding Symbols, Ranking and Analytical Reasoning.[K1]
- CO5: Investigate the different types of logics involved in Mirror and water Images, Logical Reasoning & Arithmetical Reasoning.[K4]
- CO6: Find the common traps in the questions and errors likely to be made from the concepts of Blood Relations, Directions, Average, Clock and Calendar, Data Sufficiency, Permutations-Combinations and Probability.[K3]

Text Books:

T1: PIC-VOC

T2: VERBAL ABILITY-2

T3: WORK BOOK ON APPTITUDE

Reference Books:

- Sujani,T. ,et.al, 2016, Pic Voc, Sri Vasavi Engineering College.
- RAYMOND MURPHY, 2007, Essential Grammar in Use, Cambridge University press, 3rd University.
- Thomson, A.J. , 1997, A Practical English Grammar, Oxford Publications
- Alex, K. , 2010, Soft Skills, S.Chand Publications.
- Barrons, 2014, GRE-Barons, Galgotia Publications.

- Mohammed Muneer, 2004, CAT , Tata McGraw- Hill Education
- Helio Fred Garcia, 2012, The power of Communication, FT Press/Pearson Publications
- Work book-1 on Aptitude prepared by Training & Placement Cell, Sri Vasavi Engineering College.
- Magical Book on Quicker Maths-Tyra
- Practice Book on Quicker Maths-Kundan & Tyra
- R.S.Agarwal-Sultan Chand Publications
- R.S.Agarwal-Non Verbal Reasoning

Hyperlinks

1. <https://www.indiabix.com/>
2. <https://www.campusgate.co.in/>
3. <https://www.question paper.org/>

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	50	60
2	50	60
3	50	60
4	50	60
5	50	60
6	50	60

CO 1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	To Make students understand vocabulary and its contextual usage.	K2	1	Lecture	PPT/A.V
2		Can guess the meaning of the word through the root.	K2	1	Discussion	A.V
3		Learn Synonyms-Antonyms and understand the subtle meanings of the group words.	K2	1	Lecture	PPT/A.V

4		Students learn and use new words	K2	1	Lecture	A.V
5		Can arrange sentences in sequence cohesively as per sequence	K2	1	Lecture	PPT/A.V

CO 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Students can complete the sentence in a meaningful way.	K3	1	Lecture	PPT/A.V
2		The Format and significance of official mail is learned.	K3	1	Lecture	PPT/A.V
3		Can identify the topic sentence and construct a cohesive paragraph	K3	1	Lecture	PPT/A.V

CO 3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Can draw analogy between the words of different categories.	K4	1	Lecture	PPT/A.V
2		Comprehend the passage using different strategies.	K4	1	Lecture	PPT/A.V

CO 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Students can Identify the Next letter or Number in a correct Relation.	K1	1	Lecture	PPT/A.V
2		Students can justify the relation between words and Numbers.	K1	1	Lecture	PPT/A.V
3		Students can identifying different one from group of terms.	K1	1	Lecture	PPT/A.V
4		Students can Describe their Rank in a class or in a Competition.	K1	1	Lecture	PPT/A.V

CO 5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Can Classify the ages in a family members/ Explain the relation between numbers.	K4	1	Lecture	PPT/A.V
2		Calculate the Actual time in	K4	1	Lecture	PPT/A.V

		Mirror and Water/ Classify the Images.				
3		Differentiate the logic behind the conclusions.	K4	1	Lecture	PPT/A.V
4		Students can Explain the logic for given problem.	K4	1	Lecture	PPT/A.V

CO 6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Students can choose the correct relation between the persons.	K3	1	Lecture	PPT/A.V
2		Students can show the correct direction.	K1	1	Lecture	PPT/A.V
3		Students can calculate the Average of data.	K1	1	Lecture	PPT/A.V
4		Students can Relate the correct day for given date and angle between two hands of a clock.	K1	2	Lecture	PPT/A.V
5		Students can Intercept data.	K1	1	Lecture	PPT/A.V
6		Students can report the Real Time Scenarios possibility..	K1	2	Lecture	PPT/A.V

Total No. of Classes:26

Technical Skills-III

Academic Year : 2020-21

Semester: V Sem

Branch: CSE

Programme:B.Tech

Sections:A,B,C& D

Name of the Course: Technical Skills-III

Course Code: V18CST62

COURSE OUTCOMES:

Course Outcomes (Along with Knowledge Level):

After completing this course the student able to:

CO	Course Outcomes	Knowledge Level
1	Apply fundamental data structures like List, Stack to solve real work problems in linear time i.e. $O(n)$.	K3
2	Make use of advanced data structures like queue, to solve complex problems in linear time , logarithmic time i.e. $O(n)$ or $O(n \log n)$.	K3
3	Develop programs to solve problems by with the help of searching and sorting techniques.	K3
4	Analyze linked list by comparing with Array List and develop programs to solve optimization Problems.	K4
5	Experiment with types of Linked List to solve complex combinatorial problems.	K3
6	Develop programs to solve complex problems by using combination of stack, Queue and List.	K3

TEXT BOOKS:

1. Introduction to Algorithms, Second Edition, Thomas H. Cormen Charles E. Leiserson.
2. Data Structures and Algorithms Made Easy: Narasimha Karumanchi .
3. The Algorithm Design Manual, Springer series, Steven Skiena.

Targeted Proficiency and Attainment Levels (for each course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60	60
	Level 2	50	50	50	50	50	50
	Level 1	40	40	40	40	40	40

Lecture Plan:

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO 1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs	-	-	-	
		Demonstrate ArrayList Operations.	K3	2	Lecture With Discussion	ICT
2		Illustrate Problem solving using ArrayList.	K3	6	Lecture With Discussion	ICT

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S.No	Course Outcome	Intended Learning Outcomes(ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Demonstrate Linked List Operations.	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using Linked List.	K3	6	Lecture with Discussion	ICT

08

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Demonstrate Stack Operations.	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using Stack.	K3	10	Lecture with Discussion	ICT

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S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Demonstrate Queue Operations.	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using Queue.	K4	8	Lecture with Discussion	ICT

10

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Demonstrate Searching Techniques.	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using Searching.	K3	8	Lecture with Discussion	ICT

10

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Demonstrate Sorting Techniques.	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using Sorting.	K3	8	Lecture with Discussion	ICT

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Total No. of Classes: 58